One talk a month - from symplectic to toric manifolds

derivative and integral of differential forms

$$\int f\, dx \wedge dy = \int f \boxed{?}\, \underbrace{dr \wedge d\theta}$$

$$x = r\cos\theta$$
$$y = r\sin\theta$$

$$dx = \cos\theta\, dr - r\sin\theta\, d\theta$$
$$dy = \sin\theta\, dr + r\cos\theta\, d\theta$$

$$dx \wedge dy = r\cos^2\theta\, dr \wedge d\theta + r\sin^2\theta\, dr \wedge d\theta = \boxed{r}\, dr \wedge d\theta$$

motivation

We look at the (familiar) structure of

$$\begin{array}{ccccccc} \mathcal{C}^\infty(U) & \overset{\text{grad}}{\to} & \mathrm{Vec}^\infty(U) & \overset{\text{curl}}{\to} & \mathrm{Vec}^\infty(U) & \overset{\text{div}}{\to} & \mathcal{C}^\infty(U) \end{array}$$

$$\text{curl} \circ \text{grad} = 0$$

$$\text{div} \circ \text{curl} = 0$$

where $U \subseteq \mathbb{R}^3$.

Writing this in a different notation: for $\mathcal{C}^\infty(U) \xrightarrow{\text{grad}} \mathrm{Vec}^\infty(U) \xrightarrow{\text{curl}} \mathrm{Vec}^\infty(U)$

$$f \quad \overset{\mathrm{d}}{\mapsto} \quad \mathrm{d}f = \underbrace{\textstyle\sum_i \frac{\partial f}{\partial x_i} \mathrm{d}x_i}_{\text{this looks like grad}}$$

$$\omega = \textstyle\sum_i \omega_i \mathrm{d}x_i \qquad \overset{\mathrm{d}}{\mapsto} \quad \mathrm{d}\omega = \underbrace{\textstyle\sum_i \mathrm{d}\omega_i \wedge \mathrm{d}x_i}_{\text{this looks like curl}}$$

and then continuing for $\mathrm{Vec}^\infty(U) \xrightarrow{\text{div}} \mathcal{C}^\infty(U)$

$$\mathrm{d}\omega = \underbrace{\textstyle\sum_i \mathrm{d}\omega_i \wedge \mathrm{d}x_i}_{\text{this looks like curl}}$$

$$\alpha = \textstyle\sum_{i>j} \alpha_{ij} \mathrm{d}x_i \wedge \mathrm{d}x_j \quad \overset{\mathrm{d}}{\mapsto} \quad \mathrm{d}\alpha = \underbrace{\textstyle\sum_{i>j} \mathrm{d}\alpha_{ij} \wedge \mathrm{d}x_i \wedge \mathrm{d}x_j}_{\text{this looks like div (check!)}}$$

$$\beta = \beta_{123}\, \mathrm{d}x_1 \wedge \mathrm{d}x_2 \wedge \mathrm{d}x_3 \qquad \overset{\mathrm{d}}{\mapsto} 0$$

where we understand that the "wedge" $\wedge$ works like the cross product

$$\mathrm{d}x_i \wedge \mathrm{d}x_j = -\mathrm{d}x_j \wedge \mathrm{d}x_i$$

It is easily seen

$$\mathrm{d}(\mathrm{d}(-)) = 0$$

doesn't matter what's inside the $(-)$, a real function f or the objects ω, α or β (called 0,1,2,3-forms respectively) when their components are differentiable functions.

We see this "exterior derivative"

$$\mathrm{d}$$

thus "unifies" grad, curl and div and generalizes because it may be defined as

$$\mathrm{d}\omega = \mathrm{d}\Big(\sum \omega_I \mathrm{d}x_I\Big) := \mathrm{d}\omega_I \wedge \mathrm{d}x_I$$

in any dimension.

But what are these

$$\mathrm{d}x_i$$

mean?

derivative gives you chain complex

$$\omega = \sum \omega_i \, dx_i \leftarrow \text{1-form}$$

$$d\omega = \sum d\omega_i \wedge dx_i$$

$$= \sum_{\{i,j\}}\sum \frac{\partial \omega_i}{\partial x_j} dx_j \wedge dx_i$$

$$= \sum_{(j,i)} \underbrace{\left(\frac{\partial \omega_i}{\partial x_j} - \frac{\partial \omega_j}{\partial x_i} \right) dx_j \wedge dx_i}$$

$$0 \to \text{loc const} \xrightarrow{d} \Omega^0 \xrightarrow{d} \Omega^1 \xrightarrow{d} \Omega^2 \xrightarrow{d} \cdots \to \Omega^n \xrightarrow{d} 0$$

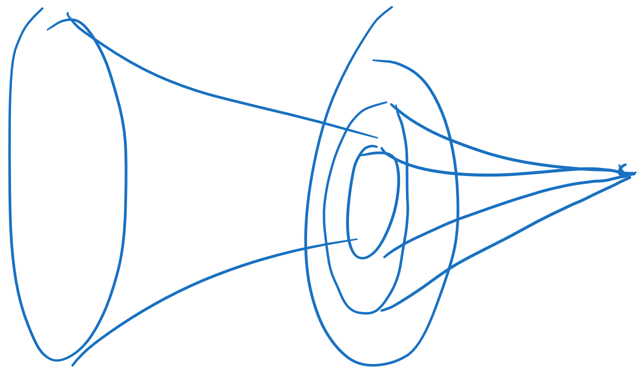
$$f \longmapsto \sum \frac{\partial f}{\partial x_i} dx_i \longrightarrow 0$$

$$\sum_n \omega_i dx_i \longrightarrow 0$$

$$\sum \alpha_{ij} \underbrace{dx_i \wedge dx_j}_{{}^nC_2}$$

$$\underbrace{\lambda \, dx_1 \wedge \ldots dx_n}$$

$d \circ d = 0$

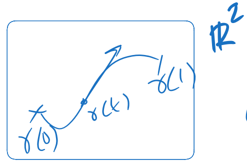


integration gives you pairing

$U \subseteq \mathbb{R}^n$

$${}^k\omega = \sum \underbrace{dx \qquad}_{k}$$

$$\boxed{\int_{[0,1]^k} \gamma^*(\omega)}$$



$$\omega = \omega_1 dx_1 + \omega_2 dx_2$$

$$:= \omega_1 \dot\gamma_1 + \omega_2 \dot\gamma_2$$

$$\gamma : [0,1] \to \mathbb{R}^2$$

$$\int_\gamma \omega := \int_{[0,1]} \underbrace{\omega_{\gamma(t)}(\dot\gamma(t))}$$

$$\dot\gamma = \begin{bmatrix}\gamma_1 \\ \gamma_2\end{bmatrix}$$

$$dx_1 = \begin{bmatrix}1 & 0\end{bmatrix}$$

$$dx_1(\dot\gamma) = \gamma_1$$

$$f : \mathbb{R}^2 \to \mathbb{R}$$

$$\int_\gamma df = f(\gamma(1)) - f(\gamma(0))$$

$$= \int_{\underbrace{\partial\gamma}} f$$

$$\int_{{}^kB} d({}^{k-1}\omega) = \int_{\underbrace{\partial^k B}} {}^{k-1}\omega$$

or $\mathbb{R}^n$

the cohomology of differential forms is topological

(de Rham isomorphism)

$$H^k(\Omega^\bullet(M), \mathrm{d}) = H^k(M, \mathbb{R})$$

flows of vector fields

motivating Hamiltonian flows

on the plane

We want a vector field that *preserves* a given smooth

$$H : \mathbb{R}^2 \to \mathbb{R}$$

Then

$$\operatorname{grad}(H)$$

does the opposite: its flow goes out of $H^{-1}(c)$ *curves*.

So we should work with the 90 degrees rotation of the gradient, so

$$J \operatorname{grad}(H)$$

where

$$J = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$$

is the clockwise 90 degrees rotation matrix.

The flow of this vector field ϕ^t such that

$$\frac{\mathrm{d}}{\mathrm{d}t}\phi^t(p) = J\mathrm{grad}(H)_{\phi^t(p)}$$

which quickly implies

$$\frac{\mathrm{d}}{\mathrm{d}t}H(\phi^t(p)) = \mathrm{d}H(\frac{\mathrm{d}}{\mathrm{d}t}\phi^t(p)) = \langle \mathrm{grad}(H), J\mathrm{grad}(H)\rangle = 0$$

by chain rule!

Hamiltonian vector fields on plane preserve area

$$X_H = \begin{bmatrix} -\frac{\partial H}{\partial y} \\ \frac{\partial H}{\partial x} \end{bmatrix}$$

$$\mathrm{div}(X_H) = -\frac{\partial}{\partial x}\frac{\partial H}{\partial y} + \frac{\partial}{\partial y}\frac{\partial H}{\partial x} = 0$$

And divergence free vector fields preserve area (volume in $\mathbb{R}^n$)

are all area preserving flows Hamiltonian vector fields?

Well,

- $X = \begin{bmatrix} X_1 \\ X_2 \end{bmatrix}$ is *Hamiltonian* for some smooth H
  - $\iff$

$$\exists H \in \mathcal{C}^\infty(\mathbb{R}^2) : \begin{bmatrix} X_1 \\ X_2 \end{bmatrix} = \begin{bmatrix} -\frac{\partial H}{\partial y} \\ \frac{\partial H}{\partial x} \end{bmatrix}$$

  - $\iff$

$$\exists H \in \mathcal{C}^\infty(\mathbb{R}^2) : \mathrm{d}H = X_2 \mathrm{d}x - X_1 \mathrm{d}y$$

- So we may use facts about 1-forms

$$X = \begin{bmatrix} X_1 \\ X_2 \end{bmatrix} \leftrightarrow \alpha = X_2 \mathrm{d}x - X_1 \mathrm{d}y$$

  - We know **on a simply connected domain** α is closed $(\mathrm{d}\alpha = 0) \iff \alpha$ is exact $\exists H \in \mathcal{C}^\infty(\mathbb{R}^2) : \alpha = \mathrm{d}H$.
  - Now

$$\mathrm{d}\alpha = \underbrace{\left(-\frac{\partial X_1}{\partial x} - \frac{\partial X_2}{\partial y}\right)}_{-\mathrm{div}(X)} \mathrm{d}x \wedge \mathrm{d}y$$

- Hence, on a **simply connected domain** flow of X is area preserving $\iff$ X is divergence free $\iff$ $\exists H \in \mathcal{C}^\infty(\mathbb{R}^2)$ such that

$$X = J\mathrm{grad}(H)$$

- **Notice a minus sign pops out in $\mathrm{d}\alpha$ above.**

isomorphism of space of symplectic modulo Hamiltonian vector fields with first cohomology

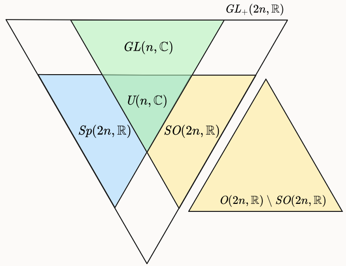
Proposition:

$$\frac{\mathrm{Sym}(M)}{\mathrm{Ham}(M)} \cong H^1(\Omega^\bullet, \mathrm{d}) \cong H^1(M, \mathbb{R})$$

summary

$\mathbb{C} = \mathbb{R}^2$	S^2	$\mathbb{C}^n = \mathbb{R}^{2n}$	(M, ω) ***symplectic manifolds***
$J = [i]$	J is the complex structure on the Riemann sphere	$J = \begin{bmatrix} i & & \\ & i & \\ & & \ddots \end{bmatrix}$	We don't need it in general!

$\mathbb{C} = \mathbb{R}^2$	S^2	$\mathbb{C}^n = \mathbb{R}^{2n}$	(M, ω) ***symplectic manifolds***
$X_H := J\mathrm{grad}(H)$	$X_H := J\mathrm{grad}(H)$	$X_H := J\mathrm{grad}(H)$	$X_H := -\omega^{-1}\mathrm{d}H$ *notice the "$-$" sign!*
$X_H H = 0$	$X_H H = 0$	$X_H H = 0$	$X_H H = 0$
area preserving $\mathcal{L}_{X_H}\mathrm{d}x \wedge \mathrm{d}y = 0$	*area preserving* $\mathcal{L}_{X_H}\lambda_{\text{area}} = 0$	volume preserving, but preserves more...	*"k-dimensional area preserving"* $\mathcal{L}_{X_H}\omega^k = 0$ for every $1 \leq k \leq n$
Converse? True for $\mathbb{R}^2$ but not true for general *open subsets!*	Converse? true for S^2!	Converse for volume preservation is false for $n > 2$, even for linear flows!	Converse is false for $\dim M > 2$! In higher dimensions Hamiltonian flows preserve ω^k for all $1 \leq k \leq n$ so preserving ω becomes a necessary condition. However, it is true that vector fields that preserve the *symplectic 2-form* are Hamiltonian on a simply connected M.

$\mathbb{C} = \mathbb{R}^2$	S^2	$\mathbb{C}^n = \mathbb{R}^{2n}$	(M, ω) ***symplectic manifolds***
Linear maps that are *generated* by Hamiltonian vector fields are just the area preserving ones $$Sp(2, \mathbb{R}) = SL(2, \mathbb{R})$$	What could be "linear" flows? well isometries $$SO(3, \mathbb{R})$$ preserve area, so we have one nice Lie group action on the 2-sphere.		symplectic G-actions, Hamiltonian G-actions
Is every area preserving flow Hamiltonian? **Yes** But not true for the space $\mathbb{R}^2 \setminus \{0\}$	Is every area preserving flow Hamiltonian? **Yes**	Is every symplectic flow Hamiltonian? **Yes**	Is every symplectic flow Hamiltonian? We have the isomorphism $$H^1(M, \mathbb{R}) \cong H^1(\Omega^\bullet, \mathrm{d}) \cong \frac{\mathrm{Sym}(M, \omega)}{\mathrm{Hamil}(M, \omega)}$$

motivating torus translations and diagonal action on $\mathbb{C}^2$

Consider the Lissajous figures:

https://www.desmos.com/calculator/8xuzb23i42

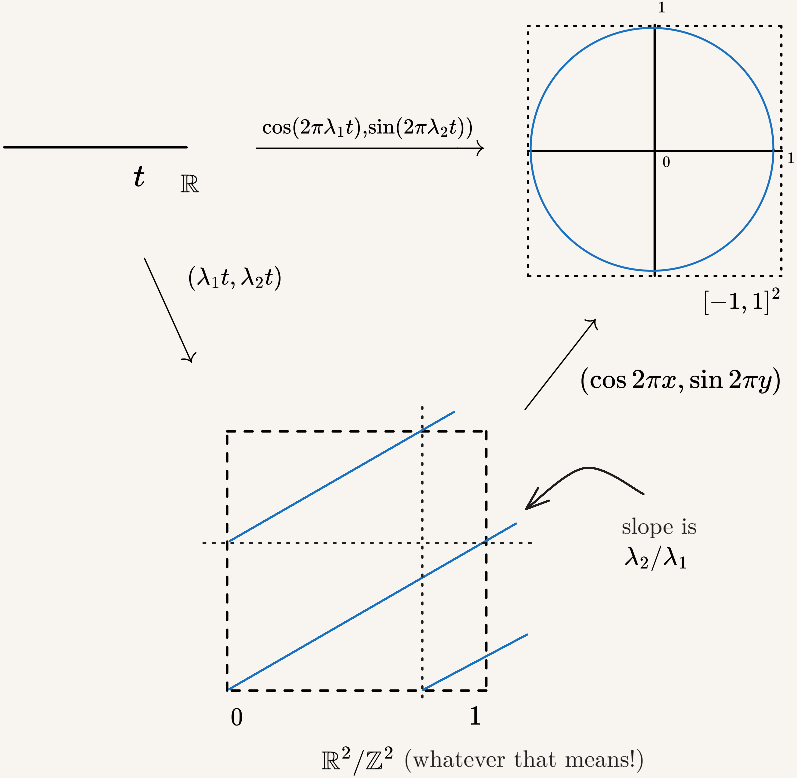
the Lissajous figure way

Proposition: Show that $\frac{\lambda_1}{\lambda_2}$ is irrational $\iff$

$$\{(\cos(2\pi\lambda_1 t), \sin(2\pi\lambda_2 t)) : t \in \mathbb{R}\}$$

is dense in $[-1,1]^2$.

simplify



Proposition:

$$\Phi := (\cos 2\pi x, \sin 2\pi y) : \mathbb{R}^2/\mathbb{Z}^2 \to [-1,1]^2$$

is surjective continuous.

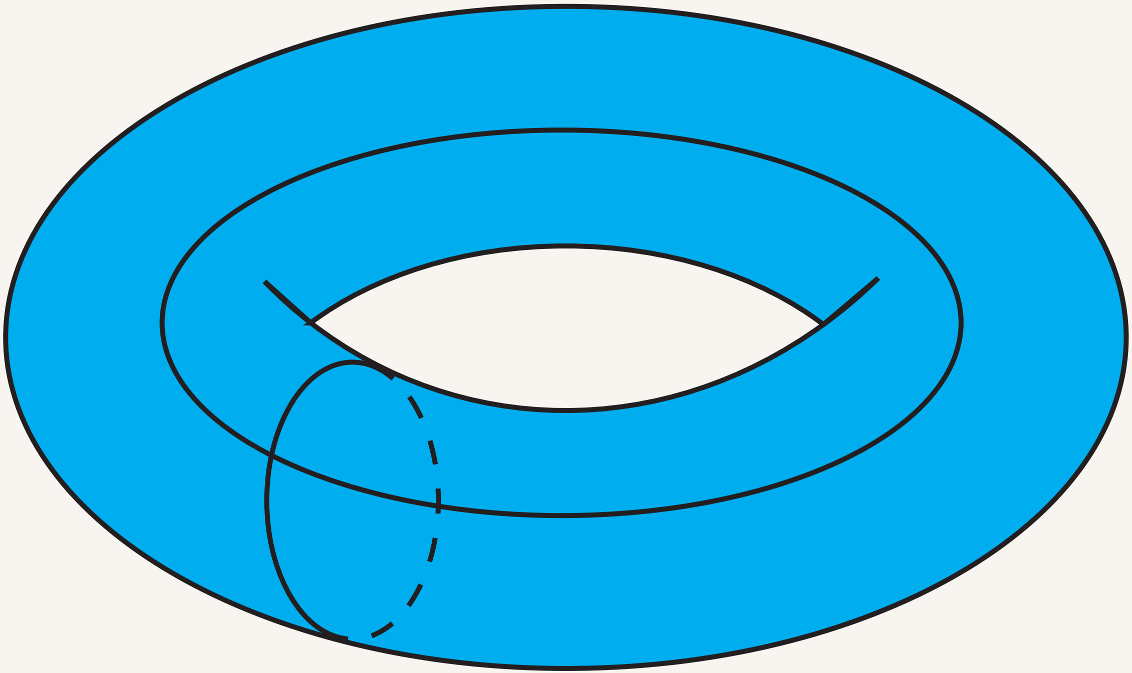
work on the simplification

Proposition:

$$\{(n\lambda_1, n\lambda_2) \bmod \mathbb{Z}^2 : n \in \mathbb{Z}\} \subset \mathbb{R}^2/\mathbb{Z}^2$$

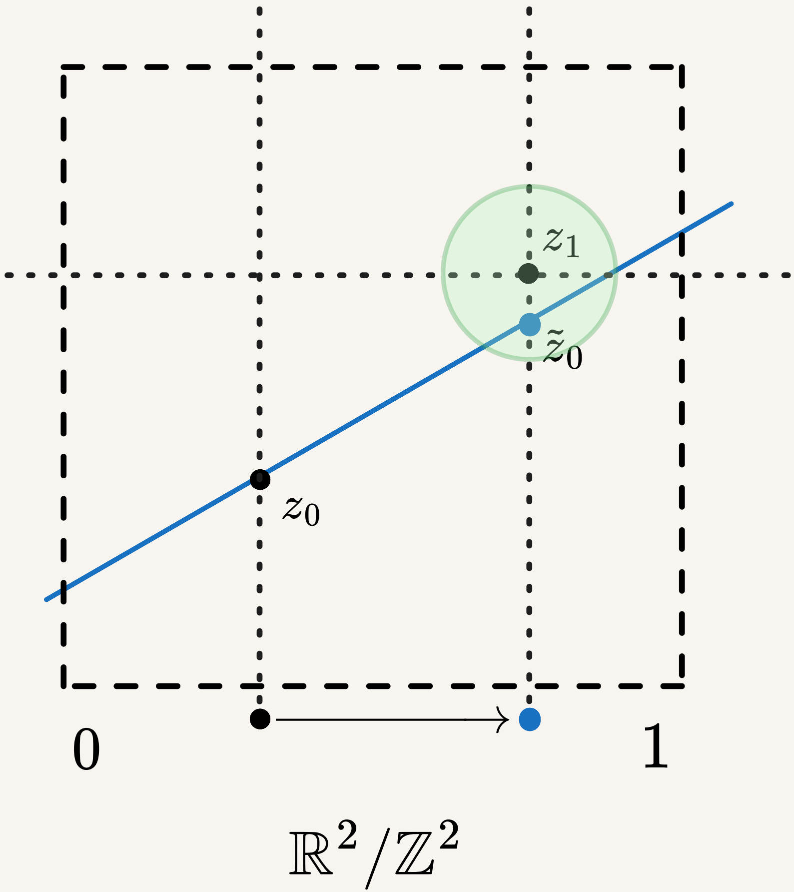
is dense in $\mathbb{R}^2/\mathbb{Z}^2$.

$\mathbb{R}^2/\mathbb{Z}^2$ is the torus!



we need to show that the "lines" on the $\mathbb{R}^2/\mathbb{Z}^2$ is a dense subset iff its slope is irrational

Start anywhere, say z_0, then the orbit of z_0 should come close to any other point z_1



A better way to look at this is

diagonal $T^2 \curvearrowright \mathbb{C}^2$

AKA the dynamics of 2 harmonic oscillators

On the symplectic 4-manifold

$$\left(\mathbb{R}^4, \sum_{i=1}^{2} \mathrm{d}p_i \wedge \mathrm{d}q_i\right) = \left(\mathbb{C}^2, \frac{1}{2i} \sum_{i=1}^{2} \mathrm{d}z_i \wedge \mathrm{d}\bar{z}_i\right)$$

we have the following Hamiltonian vector field and flow

equations	vector field	flow	Hamiltonian
$\dot p_1 = -a_1 q_1$ $\dot q_1 = a_1 p_1$ $\dot p_2 = -a_2 q_2$ $\dot q_2 = a_2 p_2$ or in more revealing, matrix form $\begin{bmatrix} \dot p_1 \\ \dot q_1 \\ \dot p_2 \\ \dot q_2 \end{bmatrix} = \begin{bmatrix} & -a_1 & \\ a_1 & & \\ & & \\ & & a_2 \end{bmatrix}$ where the matrix is a block matrix with a scaled $\pi/2$-rotation matrix	$\omega_1 \underbrace{\left(-q_1 \frac{\partial}{\partial p_1} + p_1 \frac{\partial}{\partial q_1} \right)}_{X_1} +$		$\omega_1 \underbrace{\frac{p_1^2 + q_1^2}{2}}_{H_1} + \omega_2 \underbrace{\frac{p_2^2 + q_2^2}{2}}_{H_2}$

equations	vector field	flow	Hamiltonian
$\dot z_1 = i\omega_1 z_1$ $\dot z_2 = i\omega_2 z_2$ or $\begin{bmatrix}\dot z_1\\ \dot z_2\end{bmatrix} = \begin{bmatrix} i\omega_1 & \\ & i\omega_2\end{bmatrix}\begin{bmatrix} z_1\\ z_2\end{bmatrix}$	$\omega_1 \underbrace{i z_1 \frac{\partial}{\partial z_1}}_{X_1} + \omega_2 \underbrace{i z_2 \frac{\partial}{\partial z_2}}_{X_2}$ $= \omega_1 X_1 + \omega_2 X_2$	$\begin{bmatrix} z_1\\ z_2\end{bmatrix} \overset{t}{\mapsto} \begin{bmatrix}\exp(i\omega_1 t) z_1\\ \exp(i\omega_2 t) z_2\end{bmatrix}$	$\omega_1 \underbrace{\frac{z_1 \bar z_1}{2}}_{H_1} + \omega_2 \underbrace{\frac{z_2 \bar z_2}{2}}_{H_2}$

We may rephrase in terms of the diagonal action on $\mathbb{C}^2$ by $\mathbb{T}^2$

action	generating vector field	moment map
diagonal $\mathbb{T}^2$-action on $\mathbb{C}^2$ *with both frequencies = 1* given by $\mathbb{T}^2 \curvearrowright \mathbb{C}^2$ $\begin{bmatrix} z_1\\ z_2\end{bmatrix} \xmapsto{t_1,t_2} \begin{bmatrix}\exp(it_1) z_1\\ \exp(it_2) z_2\end{bmatrix}$	the generators of the $\mathbb{T}^2$-action are the vector fields X_1, X_2 with $[X_1, X_2] = 0$	the H_1, H_2 above with $\{H_1, H_2\} = 0$ give with moment map $(H_1, H_2) : \mathbb{C}^2 \to \mathbb{R}^2$ $(z, z_2) \mapsto \frac{1}{2}(z_1 \bar z_1, z_2 \bar z_2)$ of the Hamiltonian $\mathbb{T}^2$-action

action	generating vector field	moment map
from the homomorphism $$\begin{aligned} S^1 &\to \mathbb{T}^2 \curvearrowright \mathbb{C}^2 \\ t &\mapsto l_1 t + l_2 t \end{aligned}$$ for $l_1, l_2 \in \mathbb{Z}$ we get a S^1-action on $\mathbb{C}^2$ with *integer frequencies*	the generator is $$l_1 X_1 + l_2 X_2$$	
$$\mathbb{R} \to \mathbb{T}^2$$	$$\omega_1 X_2 + \omega_2 X_2$$	
$$\mathbb{T}^2 \to \mathbb{T}^2$$		

restricting to the invariant 3-sphere

We restrict to the 3-sphere

$$S^3_h = \{(p_1, q_1, p_2, q_2) \in \mathbb{R}^4 | p_1^2 + q_1^2 + p_2^2 + q_2^2 = 2h\} \cong_{\mathsf{Man}} S^3$$

which is invariant under the flow of $\omega_1 X_1 + \omega_2 X_2$ and has orbits of $H_1 = h$. On this sphere, we have the invariant tori

$$T^2_{h_1,h_2} := \{(p_1, q_1, p_2, q_2) \in \mathbb{R}^4 | p_1^2 + q_1^2 = 2h_1, p_2^2 + q_2^2 = 2h_2\}$$

with $\omega_1 h_1 + \omega_2 h_2 = h$.

The projection

$$\begin{aligned} S^3 \setminus \{(0,0,0,1)\} &\to \mathbb{R}^3 \\ (x,y,z,w) &\mapsto (X,Y,Z) := \frac{1}{1-w}(x,y,z) \end{aligned}$$

maps

- the orbit

$$(p_1, q_1, 0, 0) \mapsto (p_1, q_1, 0)$$

  where $p_1^2 + q_1^2 = 2h_1$, the image is thus a circle in the x-y plane in $\mathbb{R}^3$
- the orbit

$$(0, 0, p_2, q_2) \mapsto \left(0, 0, \frac{p_2}{1-q_2}\right)$$

  whose image is the z-axis
- $$T^2_{h_1,h_2} \mapsto \left\{(X,Y,Z) \middle| X^2 + Y^2 = \frac{2h_1}{(1-w)^2}, Z^2 = \frac{2h_2 - w^2}{(1-w)^2}, w \neq 1\right\}$$

  which should be a torus in $\mathbb{R}^3$
-

restricting to the invariant tori: a torus translation

The subset

$$
\begin{aligned}
T^2_{a,b} &:= \{(p_1, q_1, p_2, q_2) \in \mathbb{R}^4 | p_1^2 + q_1^2 = 2a, p_2^2 + q_2^2 = 2b\} &&\subseteq (\mathbb{R}^2 \setminus \{0\})^2 \\
S^3_h &:= \{(p_1, q_1, p_2, q_2) \in \mathbb{R}^4 | p_1^2 + q_1^2 + p_2^2 + q_2^2 = 2h\} &&\subseteq \mathbb{R}^4 \setminus \{0\}
\end{aligned}
$$

which is preserved by the flow of $\omega_1 X_1 + \omega_2 X_2$.

Thus $(\mathbb{R}^2 \setminus \{0\})^2$ has a torus fibration

coordinates	X_1	$\omega_1 X_1 + \omega_2 X_2$	flow
On $\mathbb{R}^4$ we have (p_1, q_1, p_2, q_2)	$-q_1 \frac{\partial}{\partial p_1} + p_1 \frac{\partial}{\partial q_1}$	$\omega_1 \underbrace{\left(- q_1 \frac{\partial}{\partial p_1} + p_1 \frac{\partial}{\partial q_1} \right)}_{X_1}$	
On $(\mathbb{R}^2 \setminus \{0\})^2$ we have the coordinates $r_1^2 = p_1^2 + q_1^2$, θ_1, $r_2 = p_2^2 + q_2^2$, θ_2	$\frac{\partial}{\partial \theta_1}$	$\omega_1 \frac{\partial}{\partial \theta_1} + \omega_2 \frac{\partial}{\partial \theta_2}$ which is a torus translation $T^2_{a,b}$	

Flow of a general vector field

$$\omega_1 \frac{\partial}{\partial \theta_1} + \omega_2 \frac{\partial}{\partial \theta_2}$$

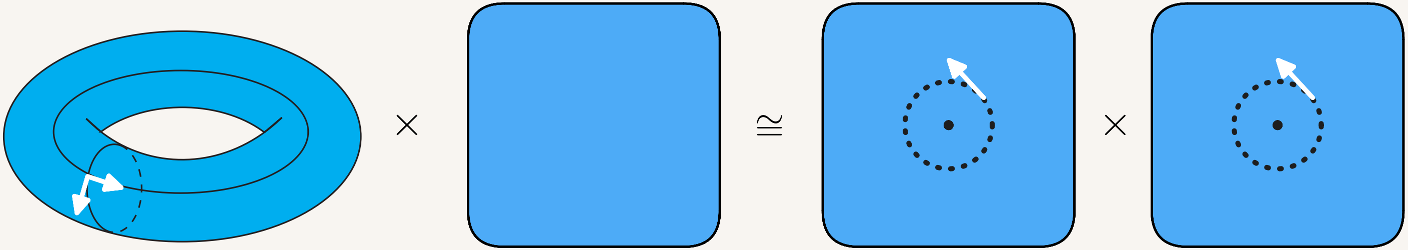
is given by

$$\exp\left(t\left(\omega_1 \frac{\partial}{\partial \theta_1} + \omega_2 \frac{\partial}{\partial \theta_2}\right)\right) \begin{bmatrix} \theta_1 \\ \theta_2 \end{bmatrix} = \begin{bmatrix} \theta_1 \\ \theta_2 \end{bmatrix} + t \begin{bmatrix} \omega_1 \\ \omega_2 \end{bmatrix}$$

whose flows have properties

if	flow is	which gives a	that is coming from
$\omega_1/\omega_2 \in \mathbb{Q}$	periodic	S^1-action on T^2	composition $$S^1 \hookrightarrow T^2 \curvearrowright T^2$$
$\omega_1/\omega_2 \in \mathbb{R} \setminus \mathbb{Q}$	not periodic, minimal (orbits are dense)	$\mathbb{R}$-action on T^2	composition $$\mathbb{R} \hookrightarrow T^2 \curvearrowright T^2$$

decomposition into torus translation $\times$ trivial

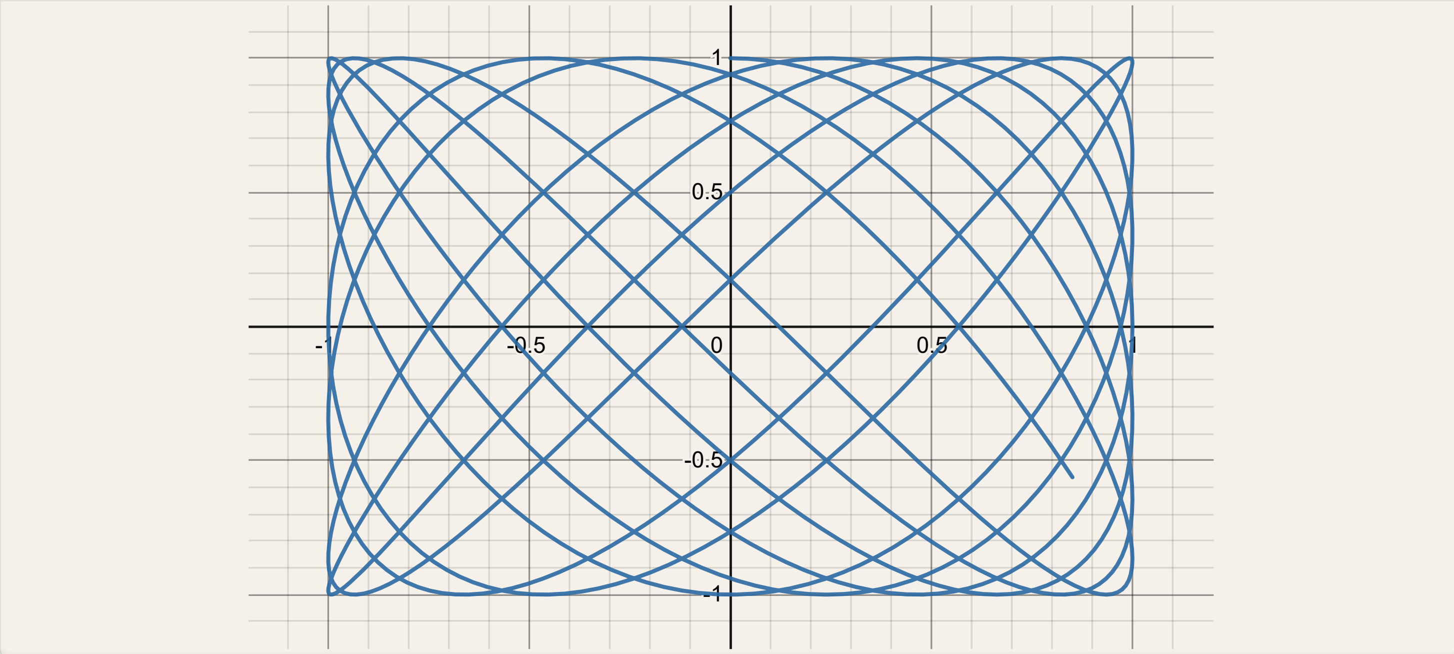


$$
\begin{array}{cccccc}
\dfrac{\partial}{\partial \theta_1} & 0 & \longmapsto & -q_1 \dfrac{\partial}{\partial p_1} + p_1 \dfrac{\partial}{\partial q_1} & 0 \\[2ex]
\dfrac{\partial}{\partial \theta_2} & 0 & \longmapsto & 0 & -q_2 \dfrac{\partial}{\partial p_2} + p_2 \dfrac{\partial}{\partial q_2}
\end{array}
$$

$$
\begin{aligned}
T^2 \curvearrowright T^2 \times \mathbb{R}^2 &\cong_{T^2} (\mathbb{R}^2 \setminus \{0\})^2 \\
v_1 \mapsto \frac{\partial}{\partial \theta_1} \times 0 &\mapsto X_1 \\
v_2 \mapsto \frac{\partial}{\partial \theta_2} \times 0 &\mapsto X_2
\end{aligned}
$$

projecting onto (q_1, q_2)

$$\{(\sin 45t, \cos 65t) | t \in [0, 1]\}$$



Projecting $(0, q_1, 0, q_2)$ we have

$$(q_1, q_2) \overset{t}{\mapsto} (\sin(\omega_1 t) q_1, \sin(\omega_2 t) q_2)$$

[1]

rational oscillator when $\omega_1, \omega_2 \in \mathbb{Z}$

A $S^1 \cong U(1)$-action (or representation) on $\mathbb{R}^4 = \mathbb{C}^2$

as a Lax pair

We define

$$L := \begin{bmatrix} p & q \\ q & -p \end{bmatrix}$$
$$M := \frac{\omega}{2}\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$$

then

$$\begin{aligned} [M, L] &= \frac{\omega}{2}\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}\begin{bmatrix} p & q \\ q & -p \end{bmatrix} - \begin{bmatrix} p & q \\ q & -p \end{bmatrix}\frac{\omega}{2}\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \\ &= \frac{\omega}{2}\begin{bmatrix} -q & p \\ p & q \end{bmatrix} - \frac{\omega}{2}\begin{bmatrix} q & -p \\ -p & -q \end{bmatrix} \\ &= \omega\begin{bmatrix} -q & p \\ p & q \end{bmatrix} \end{aligned}$$

Hence, the oscillator is equivalent to

$$\dot{L} = [M, L]$$

thus giving us conserved quantities

$$\mathrm{tr}(L^2) = \mathrm{tr}(\begin{bmatrix} p^2 + q^2 & ? \\ ? & 0 \end{bmatrix}) = p^2 + q^2$$

$$L := \begin{bmatrix} p_1 & q_1 \\ q_1 & -p_1 \end{bmatrix} + \begin{bmatrix} p_2 & q_2 \\ q_2 & -p_2 \end{bmatrix} z$$
$$M := \frac{\omega}{2} \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$$

$$[M, A_1 + A_2 z] = [M, A_1] + [M, A_2] z$$

then

$$\dot{L} = [M, L]$$

should give the equation for isotropic harmonic oscillator.

Now

$$\begin{aligned} yI_2 - A_1 - A_2 z &= \begin{bmatrix} y - p_1 - zp_2 & -q_1 - zq_2 \\ -q_1 - zq_2 & y + p_1 + zp_2 \end{bmatrix} \\ \det(yI_2 - A_1 - A_2 z) &= y^2 - (p_1 + zp_2)^2 - (q_1 + zq_2)^2 \\ &= y^2 - (p_1^2 + q_1^2) - (p_2^2 + q_2^2) z^2 - 2z(p_1 p_2 + q_1 q_2) \end{aligned}$$

which is a conic section over which

$$(y, z) \mapsto \ker(yI_2 - A(z))$$

defines a line bundle.

[2]

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1. https://www.desmos.com/calculator/irnyqunfsm ↩
2. https://www.desmos.com/calculator/2n2puejwvk ↩